

School of Advanced Computing & Information Technology

Department of Computer Science, Gujarat University

MCA123 Data Analytics (Theory+ Practical)

Assignment - Unit 4

Linear Regression

1. Given are five observations collected in a regression study on two variables.

x_i	2	6	9	13	20
y_i	7	18	9	26	23

- Develop a scatter diagram for these data.
 - What does the scatter diagram developed in part (a) indicate about the relationship between the two variables?
 - Develop the estimated regression equation for these data.
 - Use the estimated regression equation to predict the value of y when $x = 6$.
 - Compute SSE, SST, and SSR.
 - Compute the coefficient of determination r^2 . Comment on the goodness of fit.
 - Compute the sample correlation coefficient.
 - What is the value of the standard error of the estimate?
 - Test for a significant relationship by using the t test. Use $\alpha = 0.05$.
 - Use the F test to test for a significant relationship. Use $\alpha = 0.05$. What is your conclusion?
 - Develop the 95% confidence and prediction intervals when $x = 12$.
2. Given are five observations collected in a regression study on two variables.

x_i	2.6	3.4	3.6	3.2	3.5	2.9
y_i	3300	3600	4000	3500	3900	3600

- Develop a scatter diagram for these data.
- What does the scatter diagram developed in part (a) indicate about the relationship between the two variables?
- Develop the estimated regression equation for these data.
- Use the estimated regression equation to predict the value of y when $x = 4.0$.
- Compute SSE, SST, and SSR.
- Compute the coefficient of determination r^2 . Comment on the goodness of fit.
- Compute the sample correlation coefficient.
- What is the value of the standard error of the estimate?
- Test for a significant relationship by using the t test. Use $\alpha = 0.05$.

- j. Use the F test to test for a significant relationship. Use $\alpha = 0.05$. What is your conclusion?
 - k. Develop the 95% confidence and prediction intervals when $x = 5.0$.
3. Define Simple Linear Regression.
 4. Write the regression equation.
 5. What is slope in regression?
 6. What is intercept?
 7. Define dependent variable.
 8. Define independent variable.
 9. What is prediction in regression?
 10. What is residual error?
 11. What is least squares method?
 12. What is line of best fit?
 13. State any two assumptions of linear regression.
 14. What is coefficient in regression?
 15. What is regression model?
 16. What is error term?
 17. What is goodness of fit?

Practical

1. Linear Regression Model – (Predict house prices using Linear Regression)

Load **housing_price.csv** into a DataFrame.

Check for null values and drop rows containing missing data.

Separate features (X) and target ($y = \text{median_house_value}$).

Split dataset into train/test ($\text{test_size}=0.2$, $\text{random_state}=10$).

Import and train **LinearRegression()** model.

Print: R^2 score, Coefficients, Intercept

Predict on test data and compare actual vs predicted values.

Display a final summary (model accuracy, interpretation, error analysis)

2. Use FIVE separate CSV datasets and implement linear regression model.